

# DIFEI GU

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Piscataway, New Jersey, USA

## EDUCATION

### • Rutgers University

Ph.D. in Computer Science, GPA: 3.9/4.0

Sept. 2023 - Expected 2028

Piscataway, NJ, USA

◦ Advisor: [Dimitris Metaxas](#)

### • University of Waterloo

B.S. in Electrical and Computer Engineering [Co-operative Program], GPA: 3.9/4.0

Sept. 2016 - May. 2021

Waterloo, ON, Canada

## RESEARCH EXPERIENCE

### RadAlign: Advancing Radiology Report Generation with Vision–Language Concept Alignment

- Proposed a unified, concept-aligned VLM that mirrors radiologists' workflow: predict diagnostic criteria, classify diseases, and retrieve similar cases to mitigate hallucination in LLM report writing. Without needing to train the LLM itself.
- Achieved higher factual report quality (GREEN score improvement over prior work).
- Delivered strong clinical performance in sub-tasks: competitive multi-disease classification on MIMIC-CXR and robust out-of-domain generalization on IU-Xray; designed an efficient pipeline that trains the visual/concept modules while freezing the text encoder.

### Explicd: Aligning Human Knowledge with Visual Concepts for Explainable Medical Image Classification

- Introduced a simple, general framework that queries human/LLM diagnostic criteria (axes such as color, shape, texture) and aligns learnable visual concept tokens to textual "knowledge anchors."
- Trained with a criteria-anchor contrastive loss and a lightweight linear head, yielding self-explanatory predictions via per-criterion similarity scores.
- Outperformed black-box and prior explainable baselines across diverse modalities (skin lesions, colorectal histology, retinal fundus, ultrasound, chest X-ray) while retaining clear decision rationales.
- Kept the text encoder frozen for efficiency; public code and prompts enable easy extension to new tasks and datasets.

### Anatomy-VLM: A Fine-grained Vision–Language Model for Medical Interpretation

- Built an anatomy-aware *foundation model* for chest X-ray interpretation that learns transferable diagnostic priors, enabling strong zero-/few-shot performance and broad downstream gains.
- Designed queries over 29 chest regions to localize clinically meaningful areas and align each region with structured clinical phrases for fine-grained reasoning.
- Trained a multi-task objective—anatomy detection (GIoU/L1), region-specific contrastive alignment, and global disease classification—to mimic expert diagnostic workflows.
- Achieved strong zero-shot disease recognition on Chest ImaGenome and robust transfer to IU X-Ray; the learned encoder improved downstream tasks such as segmentation (e.g., heart and pneumonia masks).

## PUBLICATIONS

### Explainability & Vision–Language Alignment

- [1] Difei Gu, Yunhe Gao, Mu Zhou, Dimitris Metaxas (2025). [Anatomy-VLM: A Fine-grained Vision-Language Model for Medical Interpretation](#). In *Winter Conference on Applications of Computer Vision - WACV 2026*. [Fine-grained VLM alignment, anatomy-aware]
- [2] Difei Gu, Yunhe Gao, Yang Zhou, Mu Zhou, Dimitris Metaxas (2025). [RadAlign: Advancing Radiology Report Generation with Vision-Language Concept Alignment](#). In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025* [VLM alignment, report generation]
- [3] Yunhe Gao, Difei Gu, Mu Zhou, Dimitris Metaxas (2024). [Aligning Human Knowledge with Visual Concepts Towards Explainable Medical Image Classification](#). In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024* [Concept alignment, XAI]

### Segmentation & Reconstruction

- [1] Bangwei Guo, Yunhe Gao, Meng Ye, **Difei Gu**, Yang Zhou, Leon Axel, Dimitris Metaxas (2025). **K-Prism: A Knowledge-Guided and Prompt Integrated Universal Medical Image Segmentation Model**. In *International Conference on Learning Representations - ICLR 2026* [Universal segmentation, knowledge & prompts]
- [2] **Difei Gu**, Dexu Wang, Xiaofan Zhang, Hongsheng Li (2023). **Semi-Supervised Pulmonary Airway Segmentation with Two-Stage Feature Specialization Mechanism**. In *International symposium on biomedical imaging - ISBI 2023* [Airway CT segmentation, semi-supervised]
- [3] Changjuan Tao, **Difei Gu**, Rui Huang, Ling Zhou, Zhiqiang Hu, Yuanyuan Chen, Xiaofan Zhang, Hongsheng Li (2023). **Hippocampus segmentation after brain tumor resection via postoperative region synthesis**. In *BMC Medical Imaging* [Neuro MRI, synthesis-aided segmentation]

## PROFESSIONAL EXPERIENCE (RECENT INTERNSHIPS)

**Research Intern** · Agentic AI · Co-supervised by **Han Liu** and **Sasa Grbic** Princeton, NJ | Jun 2026 - Present  
*Siemens Healthineers – Digital Technology & Innovation (DTI)*

- Developed an **agentic autoresearch framework** for self-improving adaptation of medical foundation models, focusing on 3D chest CT diagnosis.
- Built innovative autonomous refinement loop in which LLM agents analyze model failures, propose targeted architecture and training changes, and use shared experimental memory to guide subsequent model training.
- Established a **rigorous experimentation and evaluation pipeline** with model training, controlled candidate validation, and held-out testing to ensure reproducible improvements.
- The project received best intern poster award.

**Research Intern** · Co-supervised by **Hongsheng Li** and **Xiaofan Zhang** Hong Kong SAR | Aug 2021 – May 2023  
*Centre for Perceptual and Interactive Intelligence (CPII)*

- **Hippocampus segmentation after brain tumor resection via postoperative region synthesis**. Designed VAE-GAN with *intensity alignment* and *weight-map-guided feature fusion*; jointly trained with a modified 3D SEResUNet to boost accuracy on resection cases. Achieved **+5.75 DSC points** (0.6971 → **0.7546**) and reduced **HD** from 17.2 mm to **4.68 mm** on the resection cohort; remained robust on a 2023 T1 C+ set (DSC **0.7215**, HD 6.72 mm).
- **Semi-Supervised Pulmonary Airway Segmentation with Two-Stage Feature Specialization (TFSM)**. Proposed a two-stage specialization with an *enforcement model* (main bronchi) and an SE-enhanced branch (peripheral bronchi), fused via post-processing. With only **20%** labeled CTs, improved Mean Teacher from **DSC 0.7715** to **0.8390** and HD 8.14 mm → **5.78 mm**; performance approached fully-supervised 3D U-Net (DSC 0.8657).
- Built end-to-end PyTorch pipelines (3D data I/O, augmentation, training, and evaluation), implemented SEResUNet/SE modules, and authored ablations and visualization for clinical interpretability.

**Systems Software Developer** · DevOps Ottawa, Canada | Sept 2019 – Dec 2019  
*BlackBerry*

- Actively collaborated on a large-scale automated audit tool system that tracks AWS DevOps resources.
- Integrated components with Terraform to track more than 100 cloud instances for detailed system monitoring.
- Coordinated development of automation scripts that utilize RESTful APIs to batch upload/download files for the BBLABS cloud repository.
- Led, designed, and developed a resource scheduling tool end-to-end; implemented in Python and deployed via AWS Lambda. **Guaranteed uptime and eliminated all costs from resource idling.**

**Software Development Intern** · R&D Waterloo, Canada | Jan 2019 – Apr 2019  
*Northern Digital*

- Researched and refined a critical cloud point tracking algorithm used across NDI medical applications. **Improved tracking performance by 50%.**
- Contributed to the tracking tool's CLI (C) and implemented a debugging mode.
- Designed and enhanced data visualizations for NDI's medical applications.
- Decoded real-time embedded sensor data into a custom data structure and efficiently parsed it into a GUI for realistic visual models.

## SKILLS

- **Deep Learning & Frameworks:** PyTorch, HuggingFace Transformers, CLIP, NumPy, scikit-learn
- **Vision & Multimodal:** VLM, LLM, Sparse Autoencoders, Reasoning Models, CNN/ViT
- **Optimization & Training:** Distributed Training (DDP/NCCL), Mixed Precision
- **Specialized Area:** Agentic AI, Reasoning Systems, Model Interpretability, VLM, Medical Imaging, Image Segmentation